5. Functional Specification

1. Functional Description

Business requirement:

In the current system, a PDF file is generated which contains the functional location and the material with quantity for which the replacement is needed. For this enhancement, these details must be saved in an internal table, and when the new program is run, the details of the internal table must be transferred line by line into the related task list, where the characteristics and materials in the component tab are updated accordingly.

1. Master data requirements

The following are the initial master data needed:

- Functional location classification with the characteristics indicated in section 4.1.1.2
- Task lists 1:1 with maintenance plan / item – no shared task lists, even though still defined as general task lists
- Task list or maintenance plan "tag" to indicate which characteristic is relevant for the specific maintenance plan / item. Current hypothesis to use the Inspection type field at header level of the task list
- Functional location BOM to indicate the material to be changed, with the quantity
- Class TEEXC with characteristic ELASTOMER_TYPE
- Functional Location BOM with components to be changed, assigned to the Functional Location via Construction Type field

2. Selection screen

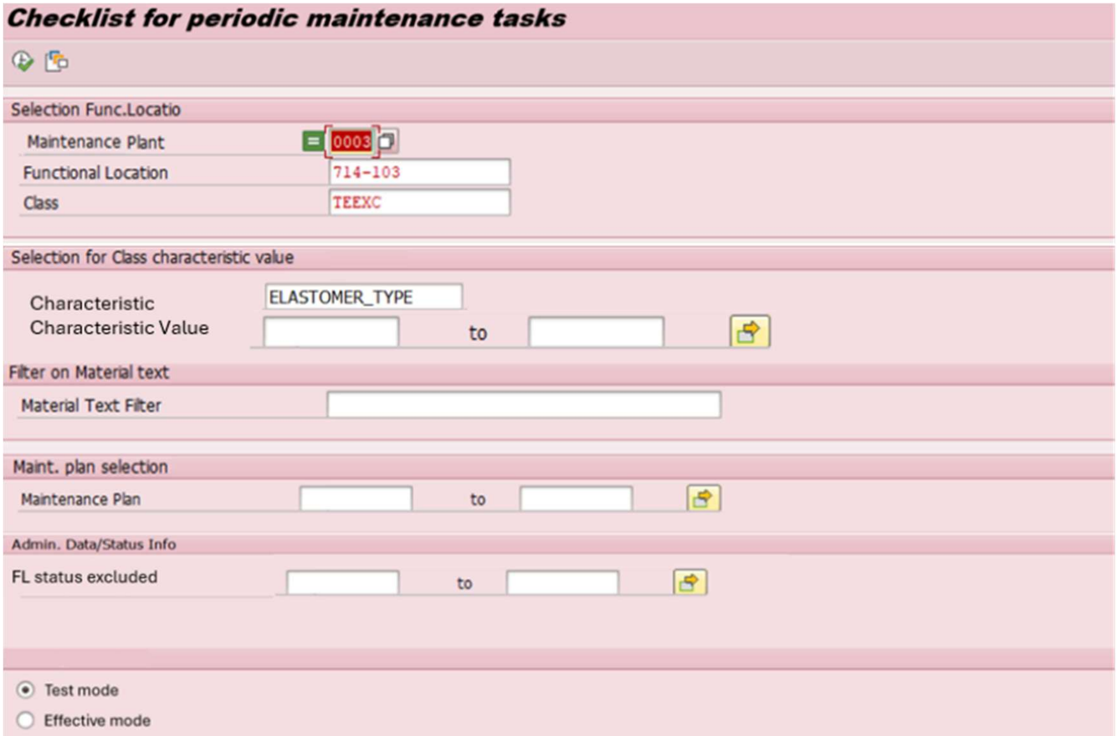

Figure 1 Proposed selection screen

1. Selection Functional Location

Table	Field	Description	Selection Type	Logic	Comment
IFLO	SWERK	Maintenance Plant		From Selection screen	Mandatory field
IFLO	TPLNR	Functional Location		From Selection screen	Mandatory field
KLAH	CLASS	Class		From Selection screen	<u>Default TEEXC</u>

		Characteristic		From Selection screen	Default ELASTOMER_TYPE

2. Selection for Class Characteristic Value

The characteristic ELASTOMER_TYPE is defaulted and at least one characteristic value from the characteristic should be chosen, else an error "Select at least one characteristic" is shown.

Characteristic Value	Description
A	Tube
B	Diaphrag
C	Seal
X	Others

3. Filter on Material Text

Table	Field	Description	Selection Type	Comment
MARA	MAKTX	Material description	Free text	

4. Maint plan selection

Table	Field	Description	Selection Type	Comment

MPLA	WARPL	Maintenance Plan	Range	

5. **Execution mode**

The users will have the option to run the program in test mode or effective mode.

The program created will be scheduled to run daily as a background job. Manual execution should be available with the option to run the program in test mode and effective mode.

- Test execution: the objects and the data to be updated would be displayed, but no actual update would be done
- Effective mode: actual update of the task lists and display of execution log

6. **Admin Data/Status Info**

The users will have the option to filter the functional locations based on their status. The statuses DLFL, INAC, OSNC, OSOT, and RETD are defaulted in the selection screen.

3. **Proposed logic**

When the program is executed, it should be able to identify and update the task lists based on the selection criteria.

A. Get the relevant functional locations:

Using the functional location entered, get all the functional locations under the FL hierarchy where the class is the class from the selection screen and the characteristics are either “Tube”, Diaphragm”, “Seal”, or “Others”. Use the Function Module “FUNC_LOCATION_HIERARCHY” to read the hierarchy using the FL’s Object ID IFLOT-

TPLNR. There should be a check on the status of the Functional Locations based on the input in the selection screen. The following statuses are defaulted to be excluded from the processing:

System Status	Description	System Value
DLFL	Deletion Flag	I0076
INAC	Object Deactivated	I0320

User Status	Description	System Value
OSNC	Out of Service - Not Calibrated	E0003
OSOT	Out of Service - Other	E0004
RETD	Retired	E0005
DCOM	Decommissioned	E0006

B. Get the material BOM information:

When the relevant functional locations have been identified, read the material BOMs linked to the functional location, and select the components together with the quantity and unit of measure.

Using the plant in the selection screen, select the materials from table MARA. If the material text filter in the selection screen is not initial, use the value to further filter the list of materials using MAKT-MAKTX.

Using the material number/s, get the material to BOM link using the table MAST where MARA-MATNR=MAST-MATNR and select the plant (MAST-WERKS), usage (MAST-STLAN), and BOM number (MAST-STLNR).

Using table BOM number, select the BOM items from table STPO where MAST-STLNR=STOP-STLNR and retrieve the component material number (STPO-IDNRK), quantity (STPO-MENGE), and unit of measure (STPO-MEINS).

The combination of the functional location, material, and quantity and unit of measure are stored in an internal table.

work in progress for the next sprint

Get the valid task lists and maintenance items:

Select the maintenance items for which the functional locations are used, and where the task list classification characteristic is either "Tube", Diaphragm", "Seal", or "Other" and where the linked maintenance plan/s is part of the selection criteria.

For each of the task lists read, select all the inspection characteristics, and compare them with the internal table above. If the entries are matching, no action is done.

Else, if the entries are different:

- *The task list existing characteristics are deleted, and replaced by the new ones, that indicate the combination of functional location, material, unit of measure and quantity*
- *The existing materials are deleted, and new materials are entered in the component tab of the task list*

.

1. Test mode

When the program is run in test mode, a view of the objects to be updated is displayed:

TEST MODE: Objects to be updated

S	Functional Location	Maintenance Plan	Maintenance item	TLType	Group	Grp.Countr	Component	Component description	Quantity	Un
	513-A6000	653419	176071	A	33765	1	06189992	Kapselelement,m.Dichtung,Spirax,BT6-...	1,000	ST
	513-A6000	604596	125427	A	30638	1	06189992	Kapselelement,m.Dichtung,Spirax,BT6-...	2,000	ST
	513-A6000	740909	267159	A	66231	1	06189992	Kapselelement,m.Dichtung,Spirax,BT6-...	1,000	ST
	513-A6000	507666	5681	A	5710	1	06189992	Kapselelement,m.Dichtung,Spirax,BT6-...	3,000	ST

Figure 2 Sample test execution log

2. Effective mode

When the program is run manually in effective mode, a log should be returned:

Elastomer change results

Msg type	Message Class	Message no	Message
S	ZI_ELSUPD	1	Task List A/XXXXX1/1 updated
S	ZI_ELSUPD	1	Task List A/XXXXX2/1 updated
S	ZI_ELSUPD	1	Task List A/XXXXX3/1 no change

Figure 3 Sample effective execution log

2. Flow Diagram

N/A

3. Unit Testing (work in progress)

Test step	Expected Result
1. Fill out the selection screen with the selection criteria and run in test mode: a. Maintenance Plant b. Functional location c. Class d. Characteristics e. Material text f. Maintenance plan	A list is displayed with the objects to be updated
2. Fill out the selection screen with the selection criteria and run in effective mode: a. Maintenance Plant b. Functional location c. Class d. Characteristics e. Material text f. Maintenance plan	The task lists are updated an execution log is displayed
3. Negative testing: Leave the fields blank and run the program	An error should be displayed "Enter selection criteria"

6. Design Specification

1. Configuration

1. Configuration reference

Configuration path **(IMG, table, ...)**	

1. Purpose of configuration

2. Workflow

1. Technical Reference

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. Flow Diagram

<<< Please provide the technical process flow diagram >>>

3. Steps Description

<<< Process steps should be descriptive in nature. The aim of the process step is to describe the overall technical process >>>

4. Technical Details

Trigger Mechanism	Mention the start condition for the workflow, e.g. on creation of a purchase document, batch program etc.
Start Condition	Example – The workflow should start only for certain document type, workflow should start only if credit amount is greater than 250000 etc.
Business Object	Mention the business object, if possible. Otherwise indicate the object in general terms (e.g., Purchase Requisition)
Standard Workflow Task / Template	In case of enhancement required for delivery workflow is required.
Level of Approval Required	
Agent Determination Technique	Role - Security Role Org Unit - HR Org Structure Custom Table - Agents in custom table Distribution Lists Unspecified - To be decided in Functional Specification Other <use to elaborate on a selection of "Other"> If the agent determination technique is different for each foreground step then please repeat this section.
Mention Logic for Agent	

Determination (if any)	
Notification Destination	Internal User (Mail Inbox) External User (email address)
Workflow Notifications Text	If any specific work item text/work item subject to be used.
Escalation Handling (if any)	If any deadline monitoring is to be done. Example: If approver does not approve for 3 business days notify his supervisor.
Integration with Portal	
Configuration Dependencies	Example setting up a new organization structure.
Error Handling (if any)	An exception situation could occur if workflow routes to a one position is vacant/not available (i.e. no user is assigned to that position.) If a specific report or additional information is required. Add attachment if necessary.
Substitution	

5. Authorization

<<< Explain which roles should be added or used to approve/reject and execute workflow items. Enter any custom authorization required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

3. Report

1. Technical Reference

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. Selection Screen Details

<<< The functional designer should be able to detail exactly what he/she wants at the selection screen merely by using this table. The programmer will be able to construct the screen directly from the details in this table. Some technical knowledge will be needed for the complete production of this table>>>

Name	Type	Parameter or Select Option	Comments (Range, Single/Multiples selections, Patterns Mandatory etc.)	Default Value
	Table-Field Check Box Radio Button with Group	Parameter Select Option		
	Table-Field	Parameter		

	Check Box Radio Button with Group	Select Option		

3. Desired Screen Design

<<< Enter attachment if necessary >>>

4. Technical Details

<<< Information like relevant database tables, data retrieval logic, type of report like (simple list report or ALV), sorting order, detail functionality, other display attributes, special interaction on clicking one or more columns etc. can mentioned here >>>

5. Starting Conditions

<<< When should the report be run? Does an interface need to be run before the report is valid, and (more commonly), should it be a batch only program (with added security) or is it needed on-line as well?

E.g. 'This program will be run after month-end billing.

E.g. 'This program will be run each time a sales order is saved >>>

6. Data Mapping Tables

<<< List of all the fields along with their details are to be mentioned here. Look and feel wise, a desired report design can also be specified here >>>

Field Name	Field Description	Output Length	Output Type	Format	Position	Screen No / Field Name

7. Report Example

<<< Use Attachment if necessary >>>

8. Authorization

<<< Explain which roles should be added or used for these reports. Enter any custom authorization if required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

4. Interface

1. Technical Reference

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. Technical Details

Interface Name	
Direction (with respect to this system)	Inbound Outbound other If other, please specify exactly
Interface Type	Batch near real-time real-time other If other, please specify exactly
Interface Frequency	Hourly Details: Daily Details: Weekly Details: Monthly Details: Quarterly Details: Yearly Details: On-Demand Details: Other Details:
Type of Records Sent	Delta Fields Delta full-record other If other, please specify exactly
Volume (per single execution)	Average Volume: <Volume> records per interface execution Peak Volume: <Lower Volume – Upper Volume>

3. Flow logic

<<< Please explain any flow logic, calculations, rules, etc.. that should be implemented in this interface >>>

4. Interface Data Layout

<<< Please list the source and destination data elements, plus any mapping that will be required for this interface. If IDOC, include segment name in structure column. Excel matching this format can be attached in place of this table. >>>

Source Structure	Source Field	Description	Data Type	Length	Transformation	Target Structure	Target Field	Description	Data Type	Length	Mand / Opt.	Comments/Remarks

5. Mapping Rules & Conversion Criteria

<<< This section should contain any additional mapping rules and conversion criteria not covered in the previous section. >>>

6. Special Case: Bi-Directional Real-Time Interface

<<< If you know this interface will be a bi-directional real-time interface (i.e. the "Source" system sends and receives data in the same execution), then a second data mapping is required. If applicable, duplicate the table from Section 4.4.4 and capture the "return data" mapping rules for the "Source" system >>>

7. Sample Data

<<< Please provide two attachments of sample source data with the expected target data after this interface is executed. Please supply the sample data in the native format or .csv, and preferably zipped >>>

8. Data Retention

<<< In file based interfaces a “backup” copy of interface data can be retained in the middleware for each execution. This can be useful for reconciliation purposes. Please indicate the retention period for this interface. If not file based, then the source or target system must fill any data retention requirements >>>

Selection		**Comments**
	None	
	7 Days	
	15 Days	
	30 Days	
	Other	

9. Middleware Solution

<<< This section should contain an outline of the chosen middleware solution and the processes involved. Middleware specific configuration should be specified >>>

10. Interface Scheduling

<<< Please describe any requirements around the timing of this interface >>>

11. Authorization

<<< Explain which roles should be created / added or users / IT for reprocessing errors or ad hoc requests. Enter any custom authorization if required. Enter the file path or folder structure to which users/IT will need access to >>>

No	Business Catalog	Authorization Parameter	Parameter Value

12. Other system documentation

<<< Reference the other system's documentation, when relevant >>>

5. Conversion

1. Technical Reference

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. Technical Details

Conversion Name	

3. Conversion Data Layout

<<< Please list the source and destination data elements, plus any mapping that will be required for this conversion. If uploading from file, source structure can be omitted. Excel matching this format can be attached in place of this table. >>>

Source Structure	Source Field	Description	Data Type	Length	Transformation	Target Structure	Target Field	Description	Data Type	Length	Mand / Opt.	Comments/Remarks

4. Mapping Rules & Conversion Criteria

<<< This section should contain any additional mapping rules and conversion criteria not covered in the previous section. >>>

5. Sample Data

<<< Please provide two attachments of sample source data with the expected target data after this conversion is executed. Please supply the sample data in the native format or .csv, and preferably zipped >>>

6. Authorization

<<< Explain which roles should be created/added or used for loading data. Enter any custom authorization if required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

6. **Enhancement**

1. **Business Add-Ins (BADIs)**

BADI Property	Value/Object
System	*<<< BTP, S/4 HANA, …>>>*
Transaction	
Enhancement Spot	
BADI Name	
Enhancement Implementation	
BADI Implementation	
Class	
Method	
Filter	
OData Service	

2. **Implicit Enhancement**

Property	Value/Object
Transaction	
Enhanced Object	
Implementation	

3. **User-Exits**

Property	Value/Object
Transaction	
Main Program	
Includes	
Form Routines	

4. **CDS Views Extension**

1. **Technical Reference**

Property	Value/Object
Original CDS View Name	
Extended CDS View Name	
Purpose of Extension	

Extension Type	☐CDS View Extension ☐Custom CDS consuming Standard CDS ☐View with Additional Associations or Joins ☐Metadata Extension
Odata Exposure	☐Yes ☐No
Input Field Parameters	
Service Definition	
Service Binding	

2. **Fields Added**

Field Name	Data Element	Source Table	Description	Annotations

5. **Function Exits**

Property	Value/Object
Transaction	
Enhancement	
Function Module Name	

Includes	

6. **Field Exits**

Property	Value/Object
Enhancement	
Main Program Name	
Function Module Name	
Field Exit Id	
Screen Number	
Screen Field Name	
Conditions for execution	

7. **Menu Exits**

Property	Value/Object
Enhancement	
Menu/Path	
Function/Transaction Code	

8. Screen Exits

Property	Value/Object
Enhancement	
Main Program Name	
Screen Number	
Program Name & Sub-Screen Number	

9. Search Help Exits

Field Name	Field Description	Import / Export (I/E)	Key Field (Y/N)	Data Element	Type (CHAR, NUMC)	Length	Default Value

10. Search Help assignment

Property	Value/Object
Standard Search Help	
Collective Search Help	
Elementary Search Help	

11. Business Transaction Events (BTE)

Property	Value/Object
Transaction	
BTE Number	
Product Name	
Function Module	

12. Custom Transaction

<<< Functional details of custom transaction can be incorporated here. Number of screens required and flow diagram can be included and provide the selection screen shot along with the table name and field name and screen shot for the required output >>>

13. Requirement routine

Menu/Submenu	
Routine number	
Business logic required	

14. Substitution

Validation Description	Fields required for validation	Point of Validation	Table used in validation	Business Rules

Substituted Field	Derived from Field	Table used in Substitution	Business Rules

15. Flow logic

<<< Please explain any flow logic, calculations, rules, etc that should be implemented in this enhancement >>>

16. Authorization

<<< Which authorization object should be used for controlled execution? Enter any custom authorization if required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

7. **Form**

1. **Technical Reference**

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. **Form Layout**

<<< Refer to the following for an output samples for Window mapping, Label Description and Field mapping >>>

C:\Documents and Settings\sutap a\My   C:\Documents and [Settings\sutapa\My D

3. **Layout Windows**

Reference	Print on page	Label Position
		X : Y :
		X : Y :
		X : Y :

		X : Y :
		X : Y :
		X : Y :
		X : Y :

4. **Field Mapping**

Field	Field Description	Functionality	Logic	Print on page	Font	Font Format	Window

5. **Standard Texts / Text Modules**

Reference	Text	Print on page	Label Position	Font	Output Format	Font Format

6. **Translation**

Reference	Description of use (in Language1)	Description of use (in Language2)	Description of use (in Language3)	Text Module Name	Notes

7. **Layout Details**

Position of Left Margin (specify unit)	
Position of Right Margin (specify unit)	
Position of Logo (specify unit)	
Logo (specify logo)	
Position of Main Window (specify unit)	

8. **Flow logic**

<<< Please explain any flow logic, calculations, rules, etc that should be implemented in this form >>>

9. **Authorization**

<<< Explain which roles should be created/added or used for printing and testing forms. Enter any custom authorization if required >>>

No	Business Catalog	Authorization Parameter	Parameter Value

8. **Fiori Application**

1. **Header Information**

Application Title	
Application ID	
Type of Enhancement	☐Custom Application ☐ Standard Application
Development Type	*<<<Fiori Elements App ....Free Style UI5 App>>>*
Application Type	*<<<List Report, Object Page , Over view Page ,etc >>>*
UI Enhancements	☐Custom Fields Added ☐UI Layout Modified ☐Extensibility Hook Used ☐Fragments or Views Introduced

2. **Technical Reference**

Object Name	Object Type	Object Description
<<< *Odata Object* >>>		
<<< *CDS View* >>>		
<<< *Custom Fields* >>>		
<<< *Catalogs* >>>		
<<< *Rules* >>>		

3. Desired Screen Design

<<< *Enter attachment if necessary* >>>

4. Technical Details

<<< *Information like relevant database tables,CDS Views,ODATA services , data retrieval logic, detail functionality, other display attributes, special interaction on clicking one or more columns etc. can mentioned here* >>>

5. Authorization

<<< *Enter Authorization Objects/fields, to be used and specific user Groups* >>>

No	Business Catalog	Name of Space (L2)	Name of Page (L3)	Name of Section (L4)	Name of App/Tile(L5)	Authorization Parameter	Parameter Value

7. Custom Tables/Structure

<<< This section should detail the attributes of any new custom table created for one of the above sections, and the properties of its fields.

NB: Existing Data Elements and/or Domains should be used whenever possible when creating custom table fields, in order to avoid unnecessary typos. In this instance, the data table row for that field should not be completed beyond 'Domain', as the remaining attributes will be default values for the selected Domain. >>>

Table Name	
Short text	
Size category	
Table maintenance allowed	
Maintenance Type	Manual / Automatic Maintenance (application table) Transportable Maintenance (customizing table)
Data class	
Buffering	
Table maintenance generation	
Authorization Group	
Change Log Enabled (Y/N)	

<table>
<tr><td>(mandatory for GxP related table)</td><td colspan="8"></td></tr>
<tr><td>SPRO Path
(mandatory for customizing tables)</td><td colspan="8"></td></tr>
<tr><td>Field Name</td><td>Data Element</td><td>Domain</td><td>Type</td><td>Length</td><td>Check Table-Field</td><td>Key Field</td><td>Foreign Key</td><td>Description</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="9">Comments</td></tr>
<tr><td colspan="9"></td></tr>
</table>

8. Error Handling

<<< Provide Error Handling details here. Job run notifications, error notifications, E-Mail messaging, custom programming, etc. may be required >>>

1. Error Messages

<<< Describe the expected error messages for different error conditions >>>

Error Message Number	**Error Message Text (70 characters)**	**Error Conditions**

9. Validation

1. Test Case References

<<< List the Test Case(s) used to validate the functionality / configuration covered in this document (IQ / OQ). >>>

Test Case ID	Test Case	Comment